RoboMix DAPT Training Data Comprehensive Survey Report

Domain Adaptive Pre-Training (DAPT) Data Survey

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Purpose: Comprehensive survey of robotics/planning-related datasets for DAPT training of the RoboMix model

Includes 83+ dataset, benchmark, and corpus sources

1. Survey Overview

This report presents the results of a comprehensive survey of datasets suitable for DAPT training of the RoboMix model. Regardless of whether the setting involves single robots or multi-robot systems, it includes all data that can be used for DAPT training, such as robot knowledge, task planning, navigation, manipulation, and hardware specifications.

Classification categories:

1. Embodied AI / task planning datasets (17)
2. Robot manipulation datasets (28; 85+ when including OXE subsets)
3. Multi-robot / HMRS planning datasets (9)
4. Robot navigation / VLN datasets (14)
5. Robot hardware specification / knowledge corpora (8 sources)
6. Large-scale synthetic datasets (5)
7. Autonomous driving / mobile-robot-related datasets (3)

2. Embodied AI / Task Planning Datasets

Datasets in which robot agents understand natural-language instructions and formulate executable task plans. In DAPT, these are core sources for learning language-action mappings and planning capabilities.

Dataset	Source/Year	Key Features	Category	Public
ALFRED	Shridhar et al. 2020	AI2-THOR; 25,743 instructions; 8,055 expert demonstrations; 428K image-action pairs; 7 task types; 120 rooms (kitchen/bedroom/bathroom/living room)	Task Planning	Public
BEHAVIOR-1K	Li et al. CVPR 2024	OmniGibson simulator; 1,000 activities; 50 scenes; 9,000+ objects; supports fluids, cloth, and non-rigid bodies; NeurIPS 2025 Challenge	Long-Horizon Planning	Public
TEACh	Padmakumar et al. AAAI-22	Amazon AI2-THOR; 3,215 human-human dialogues; separated Commander-Follower roles; household cooking tasks	Dialogue + Planning	Public

Dataset	Source/Year	Key Features	Category	Public
PARTNR	Meta 2024 (ICLR 2025)	100,000 natural-language tasks; 60 houses; 5,819 objects; four task types (free, spatial, temporal, heterogeneous); human-robot collaboration	Multi-Agent	Public
VirtualHome	Puig et al. 2018	Task graphs for home environments; RobotHow daily-activity programs; 1,000+ task scripts; ActivityGraph execution	Procedural Planning	Public
Watch-and-Help (WAH/C-WAH/O-WAH)	Puig et al. 2020	Two-agent collaborative home cleaning based on VirtualHome; includes communication (C-WAH) and online-help (O-WAH) extensions	Cooperative Planning	Public
ScienceWorld	Wang et al. 2022	Text environment covering 30 elementary-science experiment types; state changes, electricity, life science, chemistry, and genetics	Scientific Reasoning	Public
PDDLGym	Silver et al. 2020	Automatically converts PDDL to OpenAI Gym; 20+ domains such as Blocksworld, Sokoban, Minecraft, and object rearrangement	Symbolic Planning	Public
ProgPrompt	Singh et al. ICRA 2023	AI2-THOR environment; LLM-based robot task-plan generation in program form; code-as-plan approach	Code Planning	Public
Code-as-Policies	Liang et al. 2023	Converts natural language to Python programs; composes motion primitives and conditional rules; diverse robot control policies	Code Planning	Public
Instruct2Act	Huang et al. ICLR 2024	Converts multimodal instructions to Python sequences; integrates VLM-based perception, planning, and control; automatic PDDL generation	Code Planning	Public
CLEVR-Robot	Google Research	MuJoCo-based RL environment integrated with CLEVR language generation; research platform for visual-language continuous/discrete control	VL + RL Planning	Public
SMART-LLM Dataset	Kannan et al. 2023	AI2-THOR-based; 36 tasks (6 introductory, 8 simple, 14 complex, 8 advanced); evaluates LLM task decomposition, coalition formation, and assignment	Multi-Robot Planning	Public
RoCoBench (RoCo)	Mandi et al. ICRA 2024	Six multi-robot collaboration tasks; includes textual agent representations and dialectical dialogue-based collaboration	Multi-Robot Planning	Public
EB-ALFRED / EmbodiedBrain	2025	AI2-THOR; PDDL subgoal specification; discrete action set; comprehensive embodied-intelligence benchmark	Task Planning	Public

Dataset	Source/Year	Key Features	Category	Public
LoTa-Bench	Choi et al. 2024	Benchmark for language-oriented task planners; evaluates diverse LLM-based planning methods	Planning Benchmark	Public
EMMOE	2025	Comprehensive benchmark for embodied mobile manipulation in open environments; evaluates multimodal large models	Mobile Manip.	Public

2.1 DAPT Utilization Points

ALFRED: 25K+ English instructions can be used directly for language-action pair learning, with diversity across seven task types.

BEHAVIOR-1K: With 1,000 activities, it is well suited for long-horizon planning learning. The NeurIPS 2025 Challenge is ongoing.

PARTNR (Meta, ICLR 2025): 100K multi-agent collaboration tasks; the largest existing embodied multi-agent benchmark.

Code-as-Policies / ProgPrompt / Instruct2Act: Natural language -> robot-code generation paradigm. Essential data if RoboMix represents planning as code.

3. Robot Manipulation Datasets

Large-scale datasets related to object manipulation by robot arms, dual-arm systems, and humanoids. In DAPT, they support learning physical action patterns, skill selection, and tool-use methods for robots.

Dataset	Source/Year	Key Features	Public
Open X-Embodiment (OXE)	OXE Collaboration 2023	Integrates 60 datasets from 34 laboratories; 22 robot platforms; 1M+ real trajectories; used to train RT-1, RT-2, and RT-X	Public
- fractal20220817_data	Google	RT-1 demonstration data; 27.1% share of OXE	Public
- BridgeData V2	Walke et al. CoRL 2023	53,896 trajectories; 24 environments; 13 skills; natural-language labels; WidowX 250	Public
- kuka, taco_play, jaco_play, viola, etc.	Multiple	Other subsets included in OXE (60 total)	Public
DROID	Khazatsky et al. RSS 2024	76,000 demonstrations; 564 scenes; 86 tasks; 350 hours; collected by 50 people across North America, Asia, and Europe; in-the-wild	Public
RH20T	Fang et al. ICRA 2024	Comprehensive manipulation; one-shot skill learning; 40 TB (RGB-D); covers diverse tasks and skills	Public
RH20T-P	2024	Primitive-level version with fine-grained annotations; includes complex skills such as table wiping and chess	Public
RoboMIND	Wang et al. RSS 2025	107k trajectories; 479 tasks; 96 object classes; four embodiments: Franka, Tien Kung, AgileX, and UR5e	Public
AgiBot World	AgiBot + OpenDriveLab 2024	1M+ trajectories; 217 tasks; 5 scenarios; specialized for bimanual tasks; CC BY-NC-SA 4.0	Public

Dataset	Source/Year	Key Features	Public
pi0 (Physical Intelligence) Dataset	Physical Intelligence 2024	~10K hours; 68 tasks; 7 robot configurations; flow-matching-based VLA pretraining; mixed with OXE	Private
RLBench	James et al. 2020	100 unique tasks; Franka Panda; CoppeliaSim; includes language instructions	Public
Ravens / CLIPort / LoHoRavens	Zeng et al. 2021	PyBullet tabletop pick-and-place; language-conditioned policies; LoHoRavens is a long-horizon instruction extension	Public
ManiSkill2	Gu et al. 2023	20 task families; 2,000+ objects; 4M demonstrations; rigid/flexible bodies; single-arm, bimanual, and mobile settings	Public
LIBERO	Liu et al. 2023	130 language-conditioned tasks (SPATIAL/OBJECT/GOAL/100); ~20K demonstrations; Franka Panda	Public
Franka Kitchen	Gupta et al. 2019	~600 human demonstrations; seven subtasks such as microwave, light, burner, kettle, and cabinet; MuJoCo	Public
FurnitureBench	Heo et al. 2023	Real furniture assembly (cabinet, lamp, one-leg, round-table); 100 demonstrations each; FurnitureSim in parallel	Public
MetaWorld	Yu et al. 2020	50 manipulation tasks; MuJoCo; standard benchmark for multi-task and meta-RL	Public
DexGraspNet 2.0	Wang et al. CoRL 2024	ShadowHand; 1.32M grasps; 5,355 objects; cluttered scenes; 90.70% real-world success rate	Public
HOI4D	Liu et al. CVPR 2022	4D egocentric human-object interaction; 16 object categories; 50 instances; CAD models	Public
RoboAfford++	2025	Integrates six sources; object affordance recognition/prediction; spatial affordance localization; real + synthetic	Public
Ego4D	Grauman et al. CVPR 2022	3,670 hours of egocentric video; temporally dense narrations; understanding diverse everyday activities	Public
EgoExo4D	Grauman et al. CVPR 2024	1,286 hours of synchronized egocentric/exocentric video; skilled human activities; multimodal annotations	Public
RoboCodeX Dataset	Mu et al. ICML 2024	Pairs simulation tasks with execution code; multimodal reasoning pretraining; iterative self-update fine-tuning	Public
GR00T N1 Dataset	NVIDIA 2025	780K synthetic trajectories generated in 11 hours; mixed real data, internet video, and synthetic data; humanoid pretraining	Partially public
Humanoid Everyday	2024	Unitree G1/H1; Apple Vision Pro teleoperation; whole-body control including finger joints	Public

Dataset	Source/Year	Key Features	Public
RoboVerse	RSS 2025	10M+ transitions; 1,000+ tasks; MetaSim multi-simulator; synthetic data plus migrated public datasets	Public
Gemini Robotics Dataset	Google DeepMind 2025	ALOHA 2 bimanual data; SOTA on 15 benchmarks; paired with ASIMOV safety dataset	Private
ASIMOV Benchmark	Google DeepMind 2025	Robot safety; 10K safety rules; 350 real images; physical-safety and affordance categories	Public

3.1 Open X-Embodiment Sub-Datasets (All 60)

Representative list of key sub-datasets integrated into OXE (among 60 total):

1. fractal20220817_data (27.1%) - Google RT-1 Demo
2. bridge / BridgeData V2 (15.3%) - WidowX 250 multi-task data
3. kuka - KUKA robot manipulation
4. taco_play, jaco_play, roboturk, viola - diverse platforms
5. language_table - language-conditioned table manipulation (Google)
6. maniskill_dataset - ManiSkill simulation
7. furniture_bench_dataset - furniture assembly
8. stanford_hydra_dataset, stanford_kuka_multimodal_dataset
9. berkeley_autolab_ur5, berkeley_cable_routing
10. nyu_franka_play_dataset, nyu_door_opening, nyu_rot_dataset
11. columbia_cairlab_pusht_real, austin_buds_dataset, toto

Complete list of all 60 subsets: https://robotics-transformer-x.github.io/

4. Multi-Robot / HMRS Planning Datasets

Datasets related to task allocation, coalition formation, and collaborative execution in heterogeneous multi-robot systems (HMRS). This is a core target domain for RoboMix.

Dataset	Source/Year	Key Features	Public
Habitat-MAS (EMOS)	Wu et al. NeurIPS 2024	Four robot types: Fetch, Stretch, Drone, and Spot; multi-floor homes; manipulation, perception, navigation, and rearrangement; automatic capability descriptions via Robot Resume	Public
SMART-LLM Benchmark	Kannan et al. 2023	36 tasks (introductory, simple, complex, advanced); pipeline for task decomposition + coalition formation + assignment; AI2-THOR experiments	Public
RoCoBench (RoCo)	Mandi et al. ICRA 2024	Six multi-robot collaboration scenarios; includes textual agent representations	Public

Dataset	Source/Year	Key Features	Public
PARTNR Multi-Agent	Meta 2024 (ICLR 2025)	100K tasks; human-robot collaboration; four task types; based on Habitat 3.0	Public
MRTA-Sim	2025	NAV2 modular multi-robot simulator; SMT-based dynamic MRTA; indoor-delivery demonstration	Public
HMR-ODTA	2025	Online diverse-task allocation for heterogeneous mobile robot teams; adapts to dynamic scenarios	Public
CoMuRoS	2024	22 scenarios (three per task, ~20 robots); multifaceted evaluation of allocation, classification, feasibility, and accuracy	Public
MRTA Warehouse Logistics Benchmark	2024	Efficient assignment for smart-warehouse multi-delivery and heterogeneous robots; validation data for operations research	Public
Habitat 3.0 Tasks	Meta 2023	Social Navigation + Social Rearrangement; human participation via VR/keyboard; collaborative tasks	Public

4.1 Core HMRS DAPT Points

EMOS/Habitat-MAS: The most systematic heterogeneous multi-robot embodiment-aware benchmark to date. It covers collaboration among Fetch, Stretch, Drone, and Spot. The Robot Resume methodology (URDF -> automatic capability description) can be used directly in RoboMix DAPT to learn URDF-capability mappings.

PARTNR (Meta, ICLR 2025): Covers 100K-scale spatial, temporal, and heterogeneous constraints. If the human-robot collaboration data is reconstructed as robot-robot collaboration, it can be used as HMRS data.

5. Robot Navigation / Vision-and-Language Navigation (VLN) Datasets

Datasets for exploring indoor and outdoor environments using natural-language instructions or goal information. They include diverse navigation paradigms such as PointGoal, ObjectGoal, VLN, and EQA.

Dataset	Source/Year	Key Features	Public
R2R-CE (VLN-CE)	Krantz et al. 2020	Matterport3D continuous-environment VLN; 5,611 trajectories; 90 scenes; human-written English instructions; low-level control	Public
HM3D (Habitat-Matterport 3D)	Ramakrishnan et al. 2021	1,000 building-scale 3D reconstructions; 112.5k m2 exploration; supports PointGoal, ObjectGoal, and instruction navigation	Public
Matterport3D (MP3D)	Chang et al. 2017	90 indoor buildings; high-resolution RGB-D scans; standard environment backbone for R2R/VLN research	Public
Gibson Environment	Xia et al. 2018	1,400+ floors; 572 buildings; real-world-based virtualization; standard for perceptual-agent training/evaluation	Public

Dataset	Source/Year	Key Features	Public
REVERIE	Qi et al. 2020	Remote instruction navigation; 10,466 instructions (60 houses); 3,521 unseen-validation instructions; 3D object references	Public
CVDN	Thomason et al. 2019	Collaborative vision-and-dialog navigation; multiple dialogue turns; grounding-based navigation	Public
OpenEQA	Majumdar et al. CVPR 2024	1,600+ Q&A; 180+ real environments (ScanNet + HM3D); seven question categories; automatic LLM evaluation	Public
ScanQA	Azuma et al. 2022	3D spatial QA; 41k Q&A; 800 indoor scenes (ScanNet); object grounding	Public
EmbodiedScan	Wang et al. CVPR + NeurIPS 2024	5k+ scans; multimodal 3D perception; RGB-D + text; holistic 3D scene understanding	Public
SQA3D	Ma et al. 2022	Situated 3D QA; requires interpretation of agent position, orientation, and context; real-world embodied cognition	Public
HomeRobot (OVMM)	Yenamandra et al. 2023	Open-vocabulary mobile manipulation; Hello Robot Stretch; 200 3D scenes; NeurIPS 2023 Challenge	Public
MP3D-EQA	Das et al. 2018	Matterport3D-based EQA; 240k video-action pairs; links navigation and QA	Public
iGibson	Li et al. Stanford 2021	Large-scale virtualized real scenes; integrated navigation + manipulation tasks; interactive simulation	Public
MineDreamer	Zhou et al. IROS 2025	Minecraft instruction following; Chain-of-Imagination; visualizes step-by-step instruction execution	Public

6. Robot Hardware Specifications / Knowledge Corpora

Text corpora such as URDF, ontologies, datasheets, and API documentation for internalizing the physical capabilities of robots in language models, including payload, reach, speed, DOF, and sensors. These are essential for learning the ability to read URDF and automatically describe capabilities, as in EMOS Robot Resume.

Dataset/Source	Source/Year	Key Features	Public
URDF Dataset	Remos et al. 2023 (arXiv:2308.00514)	322 URDF files; 195 unique robot models; type, manufacturer, and source metadata; includes manipulators, end-effectors, quadrupeds, and wheeled robots	Public
URDD (Universal Robot Description Directory)	arXiv:2512.23135 (2024)	Structured JSON/YAML modules; extended representation of robot information derived from URDF; modular robot specifications	Public

Dataset/Source	Source/Year	Key Features	Public
IEEE RAS Robot Ontology / OCRA	IEEE RAS	Formal definitions of robot concepts and relationships; machine-interpretable representation; applied to collaborative robots	Public
Robot KG (home service)	MDPI Applied Sciences 2024	Home-environment knowledge graph; ontology-based task-processing framework; applied to KUKA YouBot	Public
Cosmos-Reason1/2 Physical Data	NVIDIA 2025	Physical commonsense + embodied reasoning data; SFT + RL post-training; 7B/8B models released on HuggingFace	Public
ASIMOV-Constitution	Google DeepMind 2025	10,000 safety rules (physical safety, affordance, etc.); annotations for 350 real images	Public
ROS2 packages / manufacturer documentation	ROS community / manufacturers	Robot APIs, action/service descriptions, NAV2, MoveIt2, and other documentation corpora; text DAPT sources	Public
Franka/UR/KUKA/Spot DataSheets, etc.	Manufacturers	Text data on payload, reach, speed, DOF, and sensor specifications; sources for verbalizing robot capabilities	Public

6.1 URDF / Specification Sources by Major Robot Platform

1. Manipulators: Franka Panda, UR3e/5e/10e/30e, KUKA LBR iiwa, ABB YuMi
2. Dual-arm robots: ALOHA 2, AgileX Dual-Arm, Rethink Robotics Baxter
3. Mobile robots: Boston Dynamics Spot, Unitree A1/Go1/G1/H1, ANYmal B/C, Clearpath Husky/Jackal
4. Humanoids: Fourier GR-1/GR-3, Apptronik Apollo, Figure 01/02, 1X NEO
5. Drones: DJI M100/Matrice 300, Skydio 2
6. Industrial AMRs: MiR100/250/500, Fetch Freight, Hello Robot Stretch

Sources: official manufacturer documentation, datasheet PDFs, ROS package READMEs, Manualsplus, robot-urdf GitHub repositories, etc.

7. Large-Scale Synthetic Datasets

Synthetic data generated in simulators to overcome the cost and safety constraints of real robot data. Co-training that mixes real and synthetic data has become a standard practice in recent VLA training.

Dataset	Source/Year	Key Features	Public
GR00T N1 Synthetic Data	NVIDIA 2025	780K synthetic trajectories (nine months of data generated in 11 hours); Isaac Sim-based whole-body RL	Public
RoboVerse Synthetic Data	RSS 2025	10M+ transitions; 1,000+ tasks; public-dataset migration + rollout + motion-planning generation	Public
NVIDIA Isaac Lab Synthetic Trajectories	NVIDIA	Humanoids such as Unitree H1 and Fourier GR-1; supports GMR real-time motion retargeting	Public

Dataset	Source/Year	Key Features	Public
EmbodiedOneVision Training Data	2025	1.2M real samples (AgiBot + OXE + RoboMIND + SO100) plus 5.7M web multimodal samples for mixed training	Partially public
InternData-A1	2024	Pioneer approach using high-fidelity synthetic data for general policy pretraining	Private

8. Autonomous Driving / Mobile-Robot-Related Datasets

Language-based planning and reasoning datasets for autonomous vehicles and mobile robots. They can be used to train high-level planning capabilities for mobile robots.

Dataset	Source/Year	Key Features	Public
nuPlan	Karnchanachari et al. 2024	1,200 hours of real-world driving (Boston, Pittsburgh, Las Vegas, Singapore); open-loop and closed-loop planning benchmark	Public
NuScenes-QA	2024	460K samples; multi-camera perception, spatial reasoning, and ego-vehicle intent QA; language-based autonomous-driving understanding	Public
DriveLM	2024	360K samples; action explanations, detailed captions, and QA pairs; LLM-based autonomous-driving planning data	Public

9. Recommended DAPT Data Selection Strategy

9.1 Priority Tier Classification

Tier 1 - Must include (HMRS / RoboMix core):

1. EMOS / Habitat-MAS: Robot Resume, heterogeneous robot collaboration, URDF-based capability descriptions
2. PARTNR: largest-scale multi-agent collaboration (100K tasks, ICLR 2025)
3. ALFRED + TEACh: task planning plus dialogue-based execution; embodied AI standards
4. OXE (Open X-Embodiment): 60 subsets, 22 robots, 1M+ trajectories
5. BEHAVIOR-1K: long-horizon planning across 1,000 activities and diverse physics simulation

Tier 2 - Strongly recommended (for domain diversity):

1. DROID + BridgeData V2: in-the-wild real manipulation with natural-language labels
2. RoboMIND + AgiBot World: large-scale multi-embodiment data
3. HM3D + VLN-CE + OpenEQA: navigation and spatial reasoning
4. SMART-LLM + RoCoBench: multi-robot task decomposition and allocation
5. Code-as-Policies + ProgPrompt + Instruct2Act: code-based planning

Tier 3 - Supplementary (for specialized capabilities):

1. DexGraspNet 2.0: dexterous grasping capability
2. HOI4D + EgoExo4D: human-object interaction and affordance learning
3. nuPlan + DriveLM: high-level planning for mobile robots
4. URDF Dataset + Robot Ontology: hardware-specification text corpora
5. GR00T / RoboVerse synthetic data: supplement data-scarce regions

9.2 Reference Data Mixing Ratios (Based on Recent Research)

1. Real robot demonstrations: ~60-70% (OXE subsets, BridgeData, DROID, RoboMIND)
2. Task planning / language annotations: ~15-20% (ALFRED, BEHAVIOR-1K, PARTNR)
3. Synthetic simulation data: ~10-15% (RoboVerse, GR00T N1 synthetic data)
4. Hardware specifications / knowledge text: ~5-10% (URDF, ontologies, datasheets)

9.3 Methods for Addressing HMRS Data Shortage

1. Generate additional data using EMOS Habitat-MAS simulation, diversifying combinations of Fetch, Stretch, Drone, and Spot
2. Expand tasks in the SMART-LLM AI2-THOR environment using diverse robot combinations
3. Reconstruct PARTNR human-robot tasks into robot-robot collaboration tasks
4. Synthesize smart-warehouse multi-robot scenarios using MRTA-Sim

10. Overall Summary

Category	Number of Datasets	Representative Datasets	DAPT Priority
Embodied AI / task planning	17	ALFRED, BEHAVIOR-1K, PARTNR, TEACh	5/5
Robot manipulation	28 (85+ including OXE subsets)	OXE, DROID, RoboMIND, AgiBot World, pi0	5/5
Multi-robot / HMRS planning	9	EMOS/Habitat-MAS, SMART-LLM, RoCo, PARTNR	5/5
Navigation / VLN / EQA	14	HM3D, VLN-CE, OpenEQA, EmbodiedScan	4/5
Hardware specifications / knowledge corpora	8 sources	URDF Dataset, Cosmos-Reason, ASIMOV	4/5
Large-scale synthetic data	5	GR00T N1, RoboVerse, EmbodiedOneVision	3/5
Autonomous driving / mobile robots	3	nuPlan, NuScenes-QA, DriveLM	3/5

Total: approximately 84+ individual dataset, benchmark, and corpus sources (excluding the 60 OXE subsets).